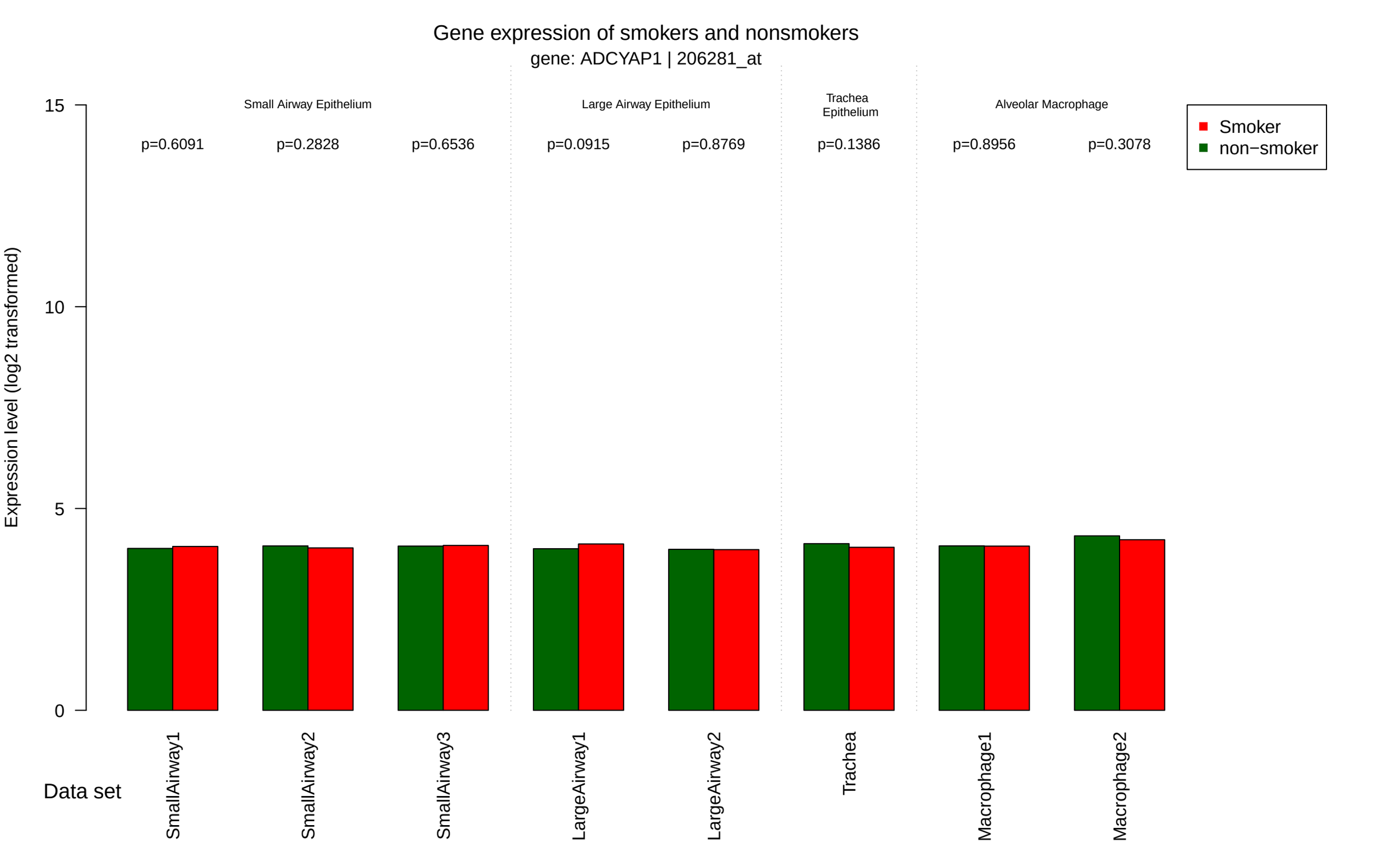
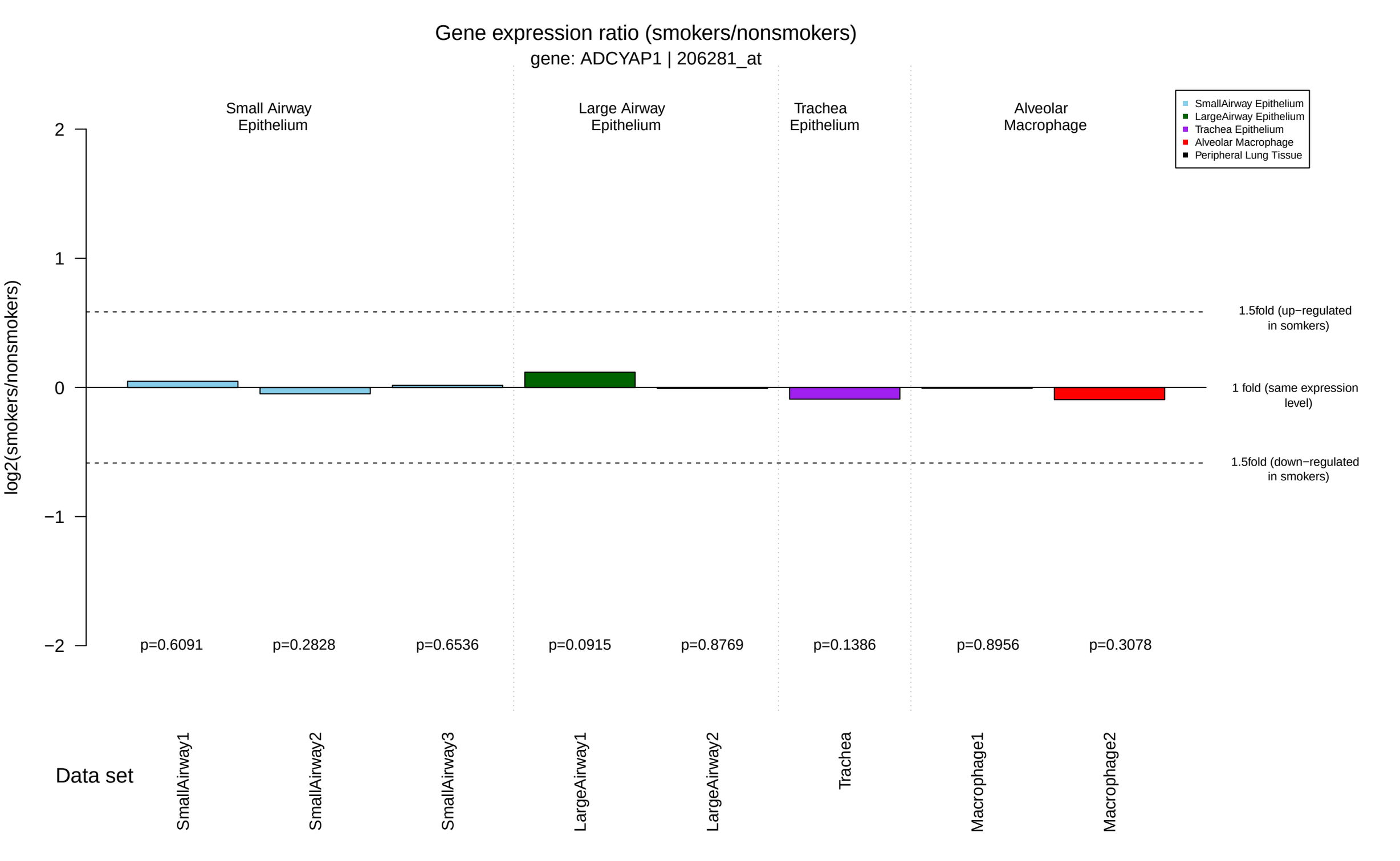
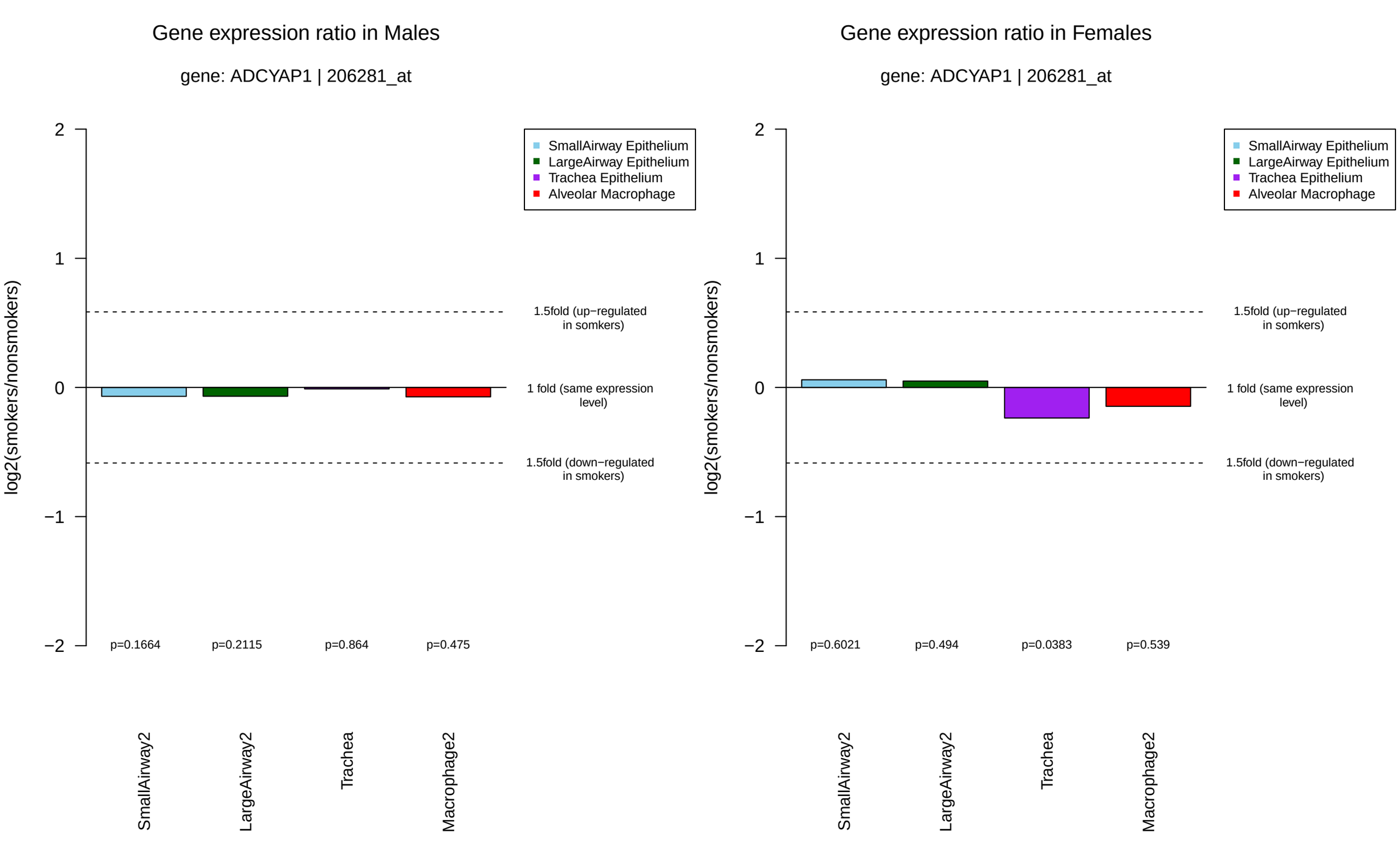
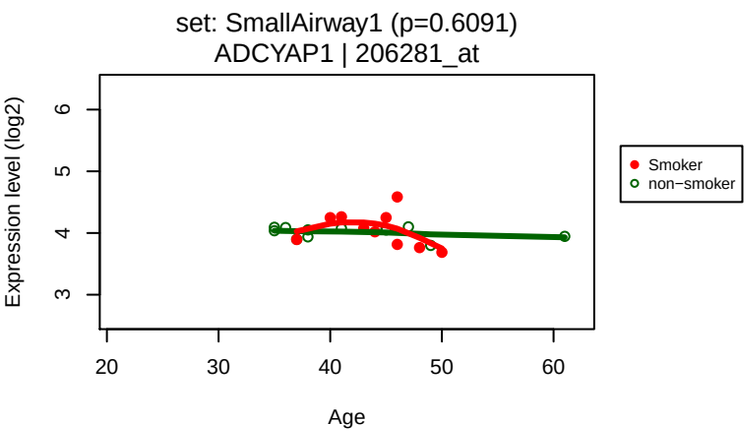
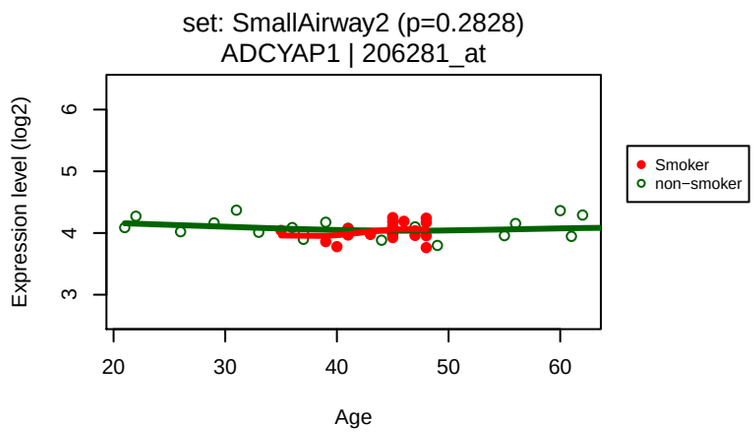
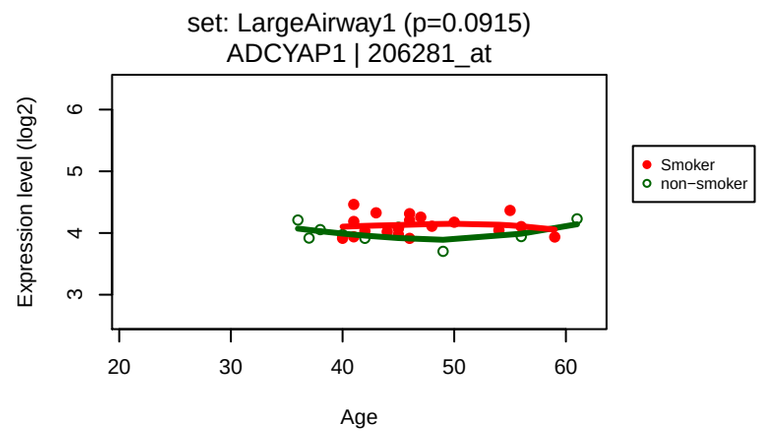

Gene expression of smokers and nonsmokers
gene: ADCYAP1 | 206281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6091
p=0.2828
p=0.6536
p=0.0915
p=0.8769
p=0.1386
p=0.8956
p=0.3078
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

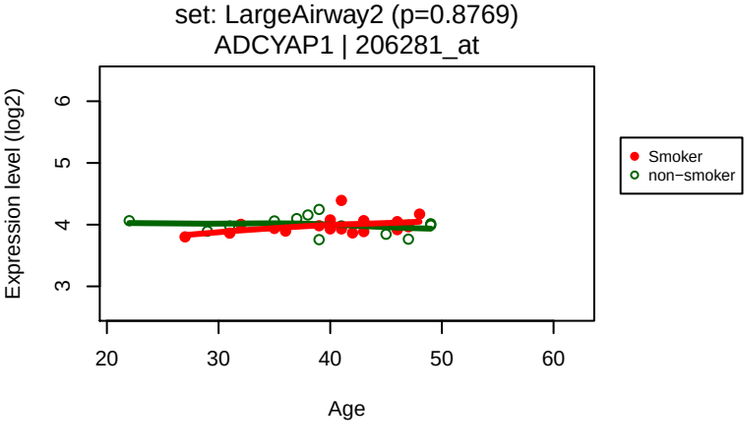

Gene expression ratio (smokers/nonsmokers)
gene: ADCYAP1 | 206281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6091
p=0.2828
p=0.6536
p=0.0915
p=0.8769
p=0.1386
p=0.8956
p=0.3078
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

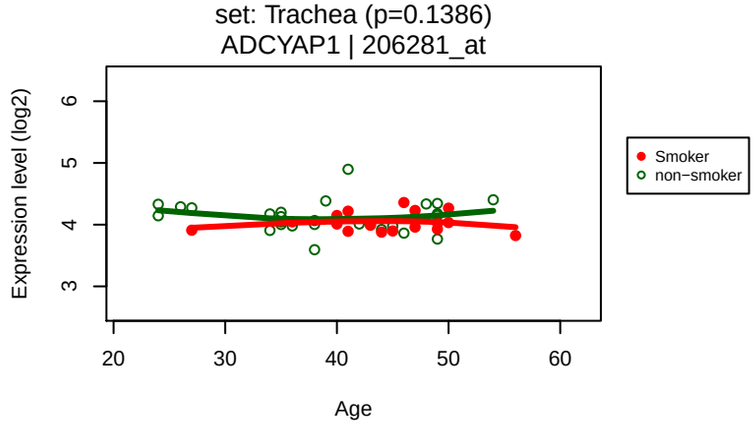

Gene expression ratio in Males
gene: ADCYAP1 | 206281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1664
p=0.2115
p=0.864
p=0.475
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCYAP1 | 206281_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6021
p=0.494
p=0.0383
p=0.539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

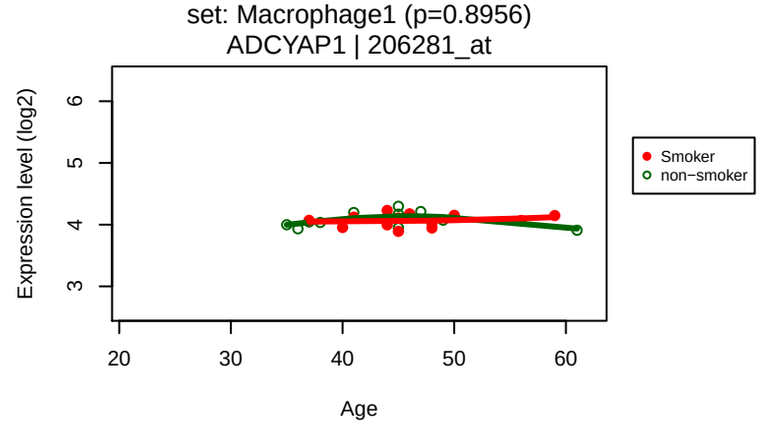

set: SmallAirway1 (p=0.6091)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker

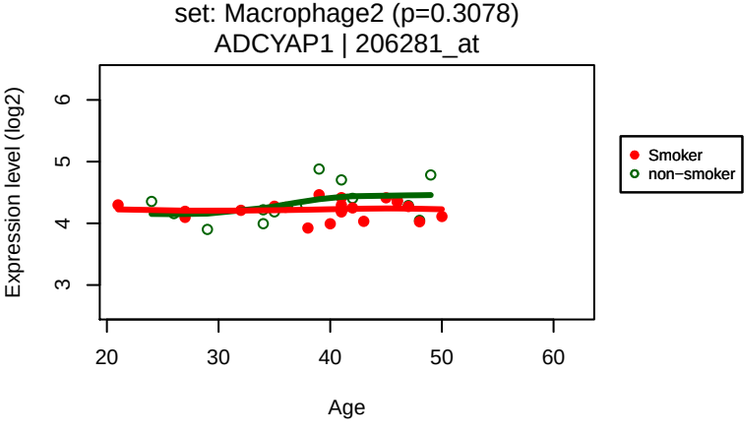

set: SmallAirway2 (p=0.2828)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker

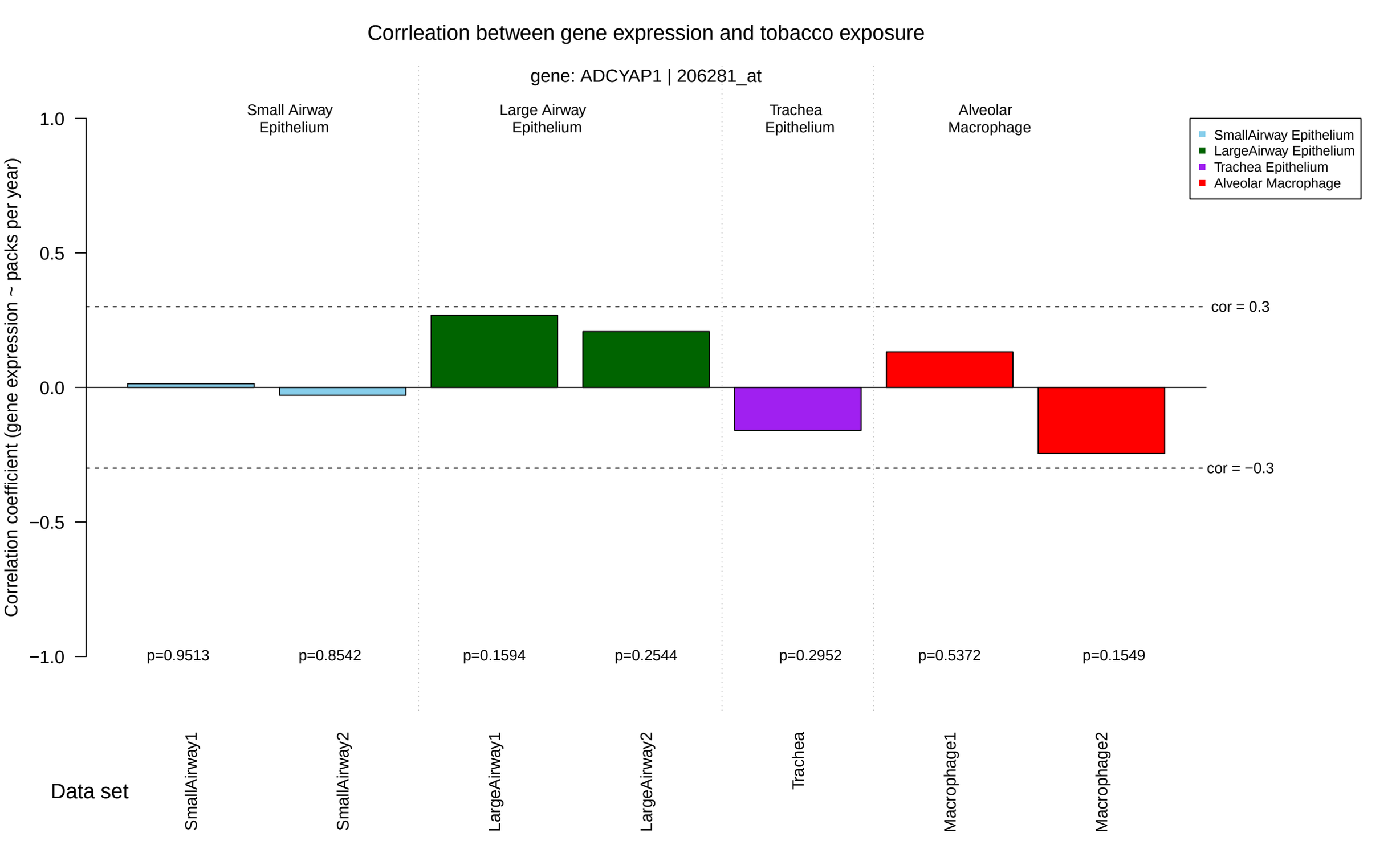

set: LargeAirway1 (p=0.0915)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8769)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1386)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8956)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3078)
ADCYAP1 | 206281_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCYAP1 | 206281_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9513
p=0.8542
p=0.1594
p=0.2544
p=0.2952
p=0.5372
p=0.1549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2